

Visuals for Determinants of Infidelity

2026-06-01

```
library(readxl)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)

setwd("~/Desktop/econometrics")

affairs <- read_xls("affairs (2).xls", col_names = FALSE)
```

```
## New names:
## • `` -> `...1`
## • `` -> `...2`
## • `` -> `...3`
## • `` -> `...4`
## • `` -> `...5`
## • `` -> `...6`
## • `` -> `...7`
## • `` -> `...8`
## • `` -> `...9`
## • `` -> `...10`
## • `` -> `...11`
## • `` -> `...12`
## • `` -> `...13`
## • `` -> `...14`
## • `` -> `...15`
## • `` -> `...16`
## • `` -> `...17`
## • `` -> `...18`
## • `` -> `...19`
```

```
colnames(affairs) <- c(
  "id", "male", "age", "yrsmarr", "kids", "relig", "educ", "occup",
```

```

"ratemarr", "naffairs", "affair", "vryhap", "hapavg", "avgmarr",
"unhap", "vryrel", "smerel", "slghtrel", "notrel"
)

affairs$affair <- ifelse(affairs$affair > 0, 1, 0)

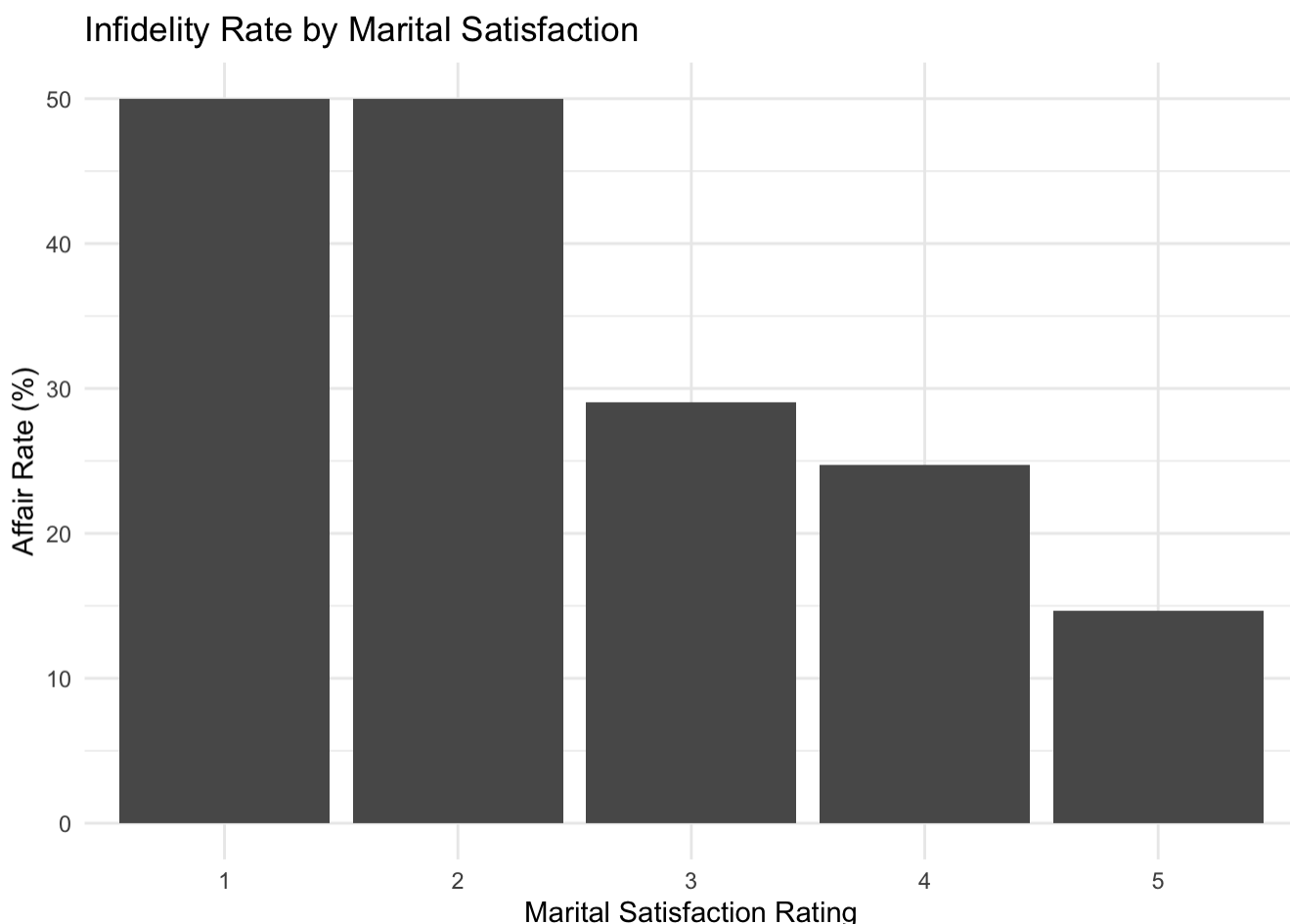
```

1. Infidelity rate by marital satisfaction

```

affairs %>%
  group_by(ratemarr) %>%
  summarise(affair_rate = mean(affair) * 100) %>%
  ggplot(aes(x = factor(ratemarr), y = affair_rate)) +
  geom_col() +
  labs(
    title = "Infidelity Rate by Marital Satisfaction",
    x = "Marital Satisfaction Rating",
    y = "Affair Rate (%)"
  ) +
  theme_minimal()

```



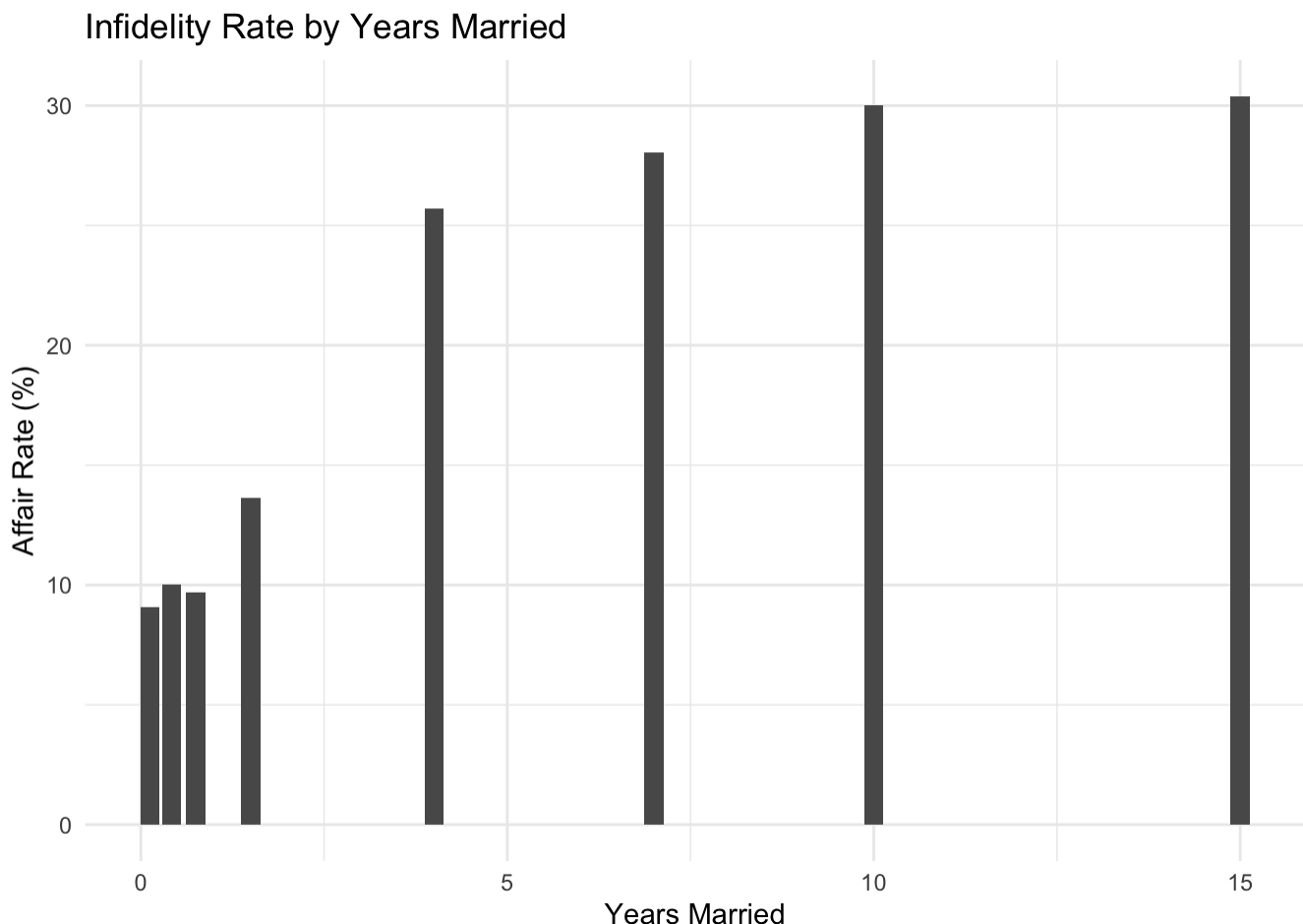
```

ggsave("marital_satisfaction_affair_rate.png", width = 8, height = 5)

```

2. Infidelity rate by years married

```
affairs %>%
  group_by(yrsmarr) %>%
  summarise(affair_rate = mean(affair) * 100) %>%
  ggplot(aes(x = yrsmarr, y = affair_rate)) +
  geom_col() +
  labs(
    title = "Infidelity Rate by Years Married",
    x = "Years Married",
    y = "Affair Rate (%)"
  ) +
  theme_minimal()
```

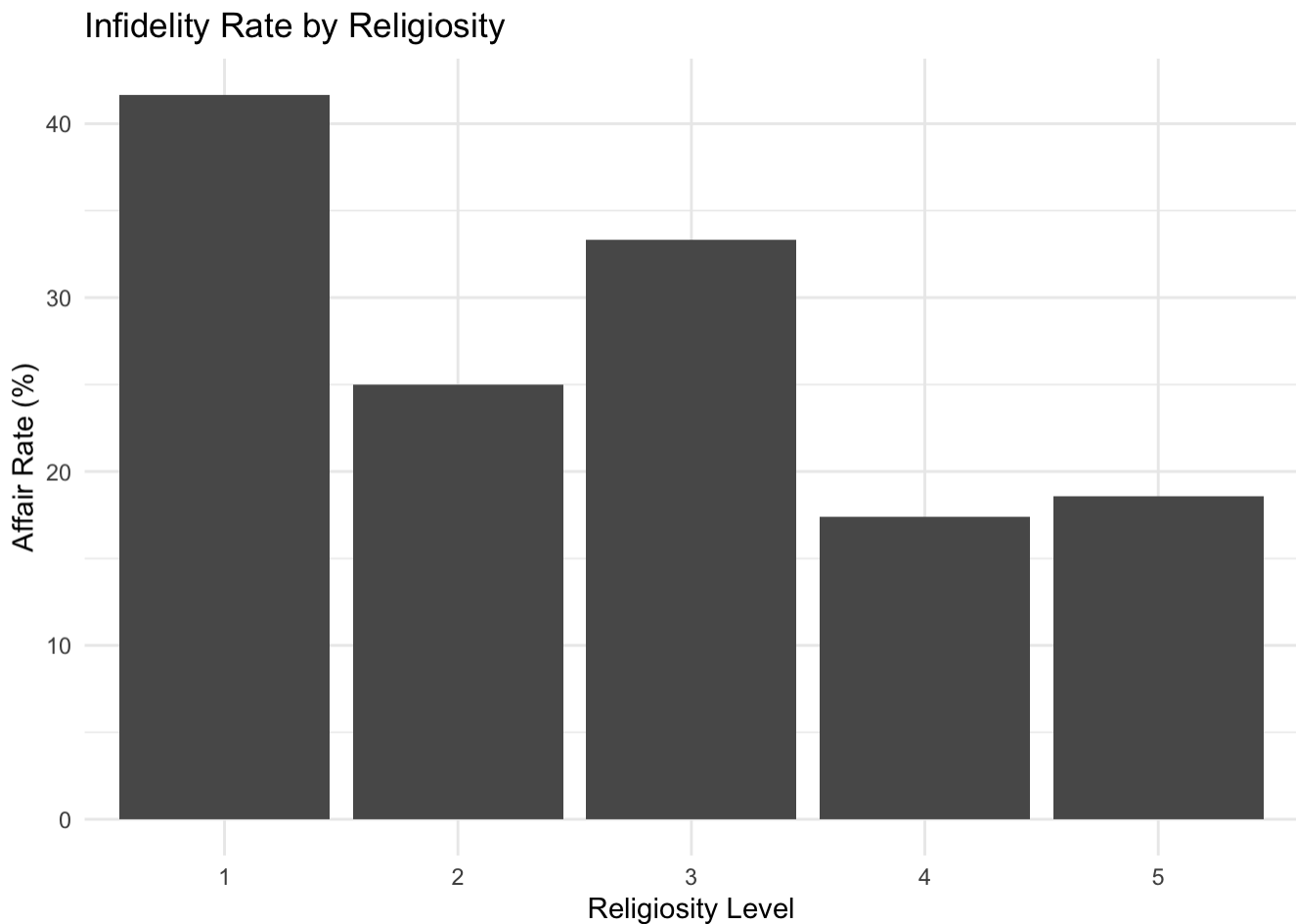


```
ggsave("years_married_affair_rate.png", width = 8, height = 5)
```

3. Infidelity rate by religiosity

```
affairs %>%
  group_by(relig) %>%
  summarise(affair_rate = mean(affair) * 100) %>%
  ggplot(aes(x = factor(relig), y = affair_rate)) +
  geom_col() +
  labs(
    title = "Infidelity Rate by Religiosity",
    x = "Religiosity Level",
```

```
y = "Affair Rate (%)"
) +
theme_minimal()
```



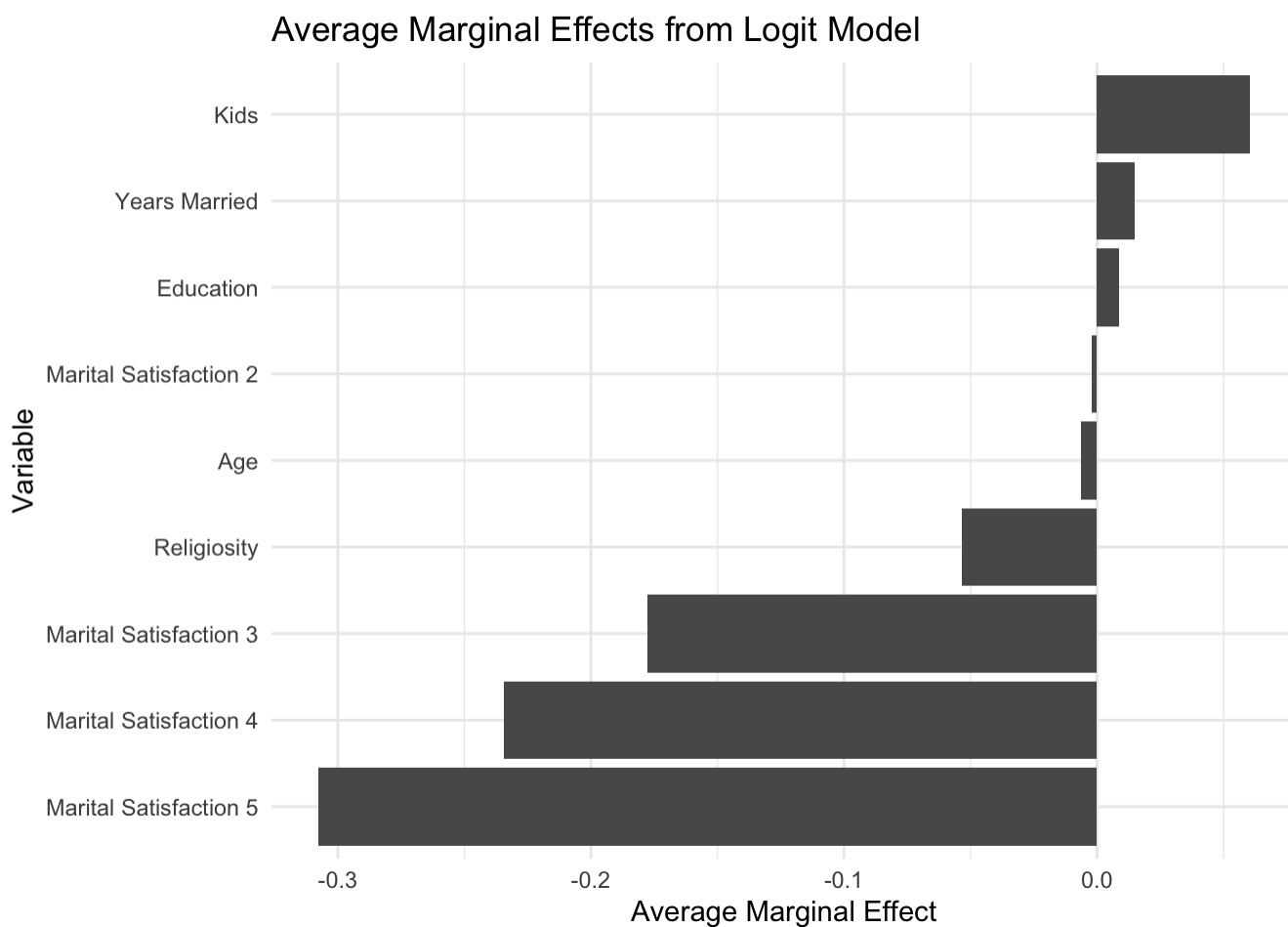
```
ggsave("religiosity_affair_rate.png", width = 8, height = 5)
```

4. Average marginal effects chart

```
marginal_effects <- data.frame(
  variable = c("Age", "Education", "Kids", "Marital Satisfaction 2",
               "Marital Satisfaction 3", "Marital Satisfaction 4",
               "Marital Satisfaction 5", "Religiosity", "Years Married"),
  ame = c(-0.0065, 0.0089, 0.0603, -0.0019,
          -0.1775, -0.2342, -0.3078, -0.0536, 0.0149)
)

ggplot(marginal_effects, aes(x = reorder(variable, ame), y = ame)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Average Marginal Effects from Logit Model",
    x = "Variable",
    y = "Average Marginal Effect"
```

```
) +  
theme_minimal()
```



```
ggsave("marginal_effects_chart.png", width = 8, height = 5)
```